

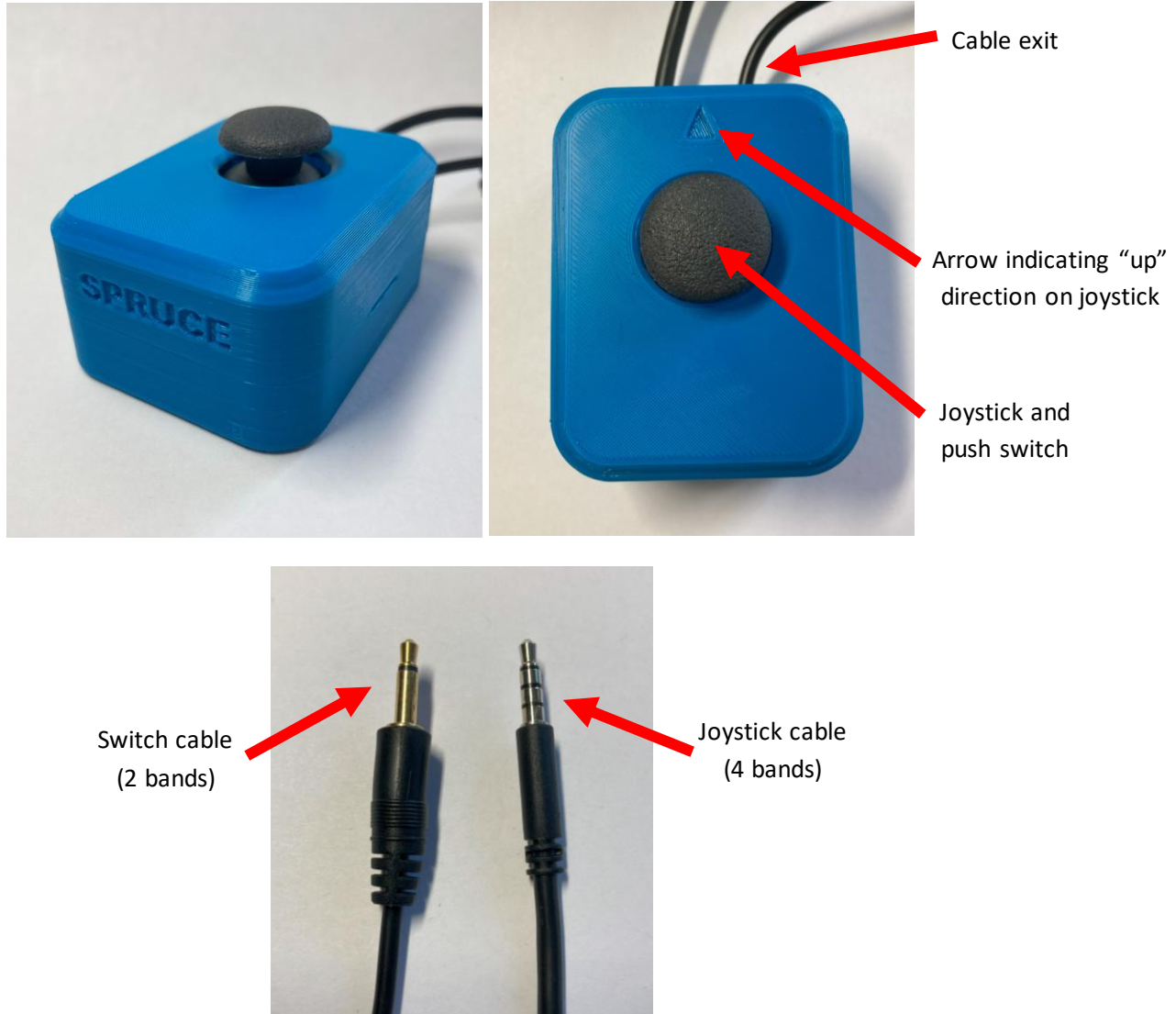
Spruce Mini Joystick

USER QUICK GUIDE

Introduction

The Spruce Mini Joystick is a small analog joystick compatible with the XAC or other analog input joystick devices, such as the Forest Joystick Mouse Hub. It has an included switch, that is activated by pressing down on the joystick topper.

Features



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Usage

Using the Joystick Alone with Host Device

1. Plug the 3.5 mm TRRS cable (4 bands) from the joystick into the host device analog joystick input (such as the Xbox Adaptive Controller). If using the Xbox Adaptive Controller, plug it into X1 to use at the left joystick, or plug into X2 to use as the right joystick.
2. Ensure the arrow on the joystick is pointing in the “up/forward” direction, away from the user.
3. Mount the device if needed (see below).
4. Move joystick as you would with a standard controller (forward, left, backward, right).

Using the Joystick and Push Switch with Host Device

1. Plug the switch cable (2 bands) into the desired switch input. If using the Xbox Adaptive Controller, this could be any port other than the X1 and X2 ports.
2. Plug the 3.5 mm TRRS cable (4 bands) from the joystick into the host device analog joystick input (such as the Xbox Adaptive Controller). If using the Xbox Adaptive Controller, plug it into X1 to use at the left joystick, or plug into X2 to use as the right joystick.
3. Ensure the arrow on the joystick is pointing in the “up/forward” direction, away from the user.
4. Mount the device if needed (see below).
5. Move joystick as you would with a standard controller (forward, left, backward, right).

Note: If using the Xbox Adaptive Controller (XAC), and using the push switch, **make sure to plug in the switch cable before the joystick cable**, or before moving the joystick. If the joystick was plugged in first and moved around, please power the XAC on and off before plugging in both the switch and joystick.

If the switch cable was previously plugged in, but now the joystick is being used alone, power the XAC off and on again.

Compatibility

This device is compatible with the [Xbox Adaptive Controller \(XAC\)](#), [PlayStation Access Controller](#), [Forest Hub](#), [Enabled Controller](#), [Enabled Controller Mini](#), and any other device that accepts a TRRS analog joystick input.

Drift Troubleshooting

If you experience drift with the Spruce Mini Joystick when using with the Xbox Adaptive Controller:

1. Unplug the Xbox Adaptive Controller (XAC) if connected via USB.
2. Power the XAC off.
3. Plug the XAC back in if connecting via USB, or power on and connect via Bluetooth.
4. Move the joystick around in circles.

This process should resolve drift issues. Pay attention to the order that the joystick and switch cables are plugged in, if using the switch always plug it in first. If using the joystick alone do not plug in the switch.



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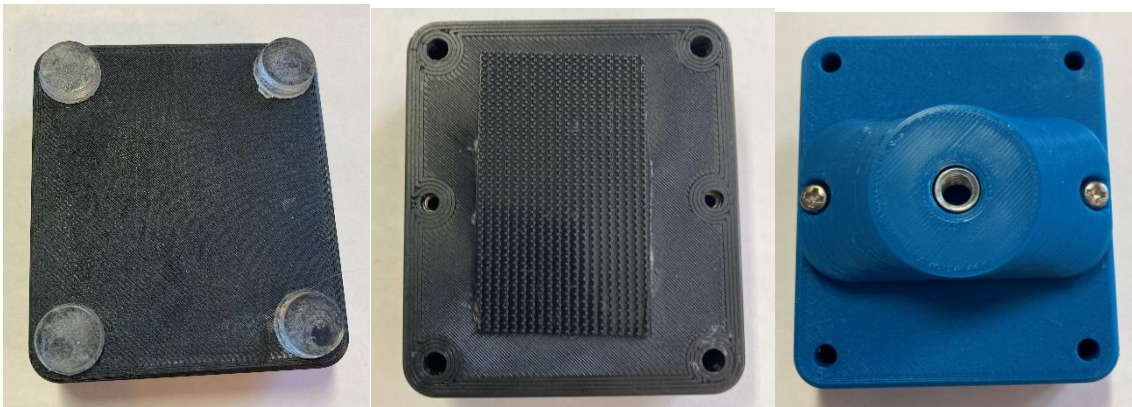
Mounting the Joystick

The Spruce Mini Joystick can be mounted using various methods, depending on your preference and how you can best access it. See the [OpenAT Joystick Setup Guide](#) for more information on different ways to mount the joystick.

A few options are:

1. Tabletop
 - a. Alone
 - b. With non-slip pads
 - c. With hook and loop fasteners (such as Velcro)
2. Using a mount adapter
 - a. Camera mount adapter used with camera mount, or any mount with ¼-20 threads

Note: The mounting options are shown on a different joystick but will apply the same to this one.



Changing Toppers

The Spruce Joystick comes with a standard rubber like small dome topper like shown in the feature photo above. There are various 3D printed toppers that are available for the Spruce Joystick that can replace the standard topper. Such as the following:



To add a 3D printed topper on the Spruce, follow these instructions:

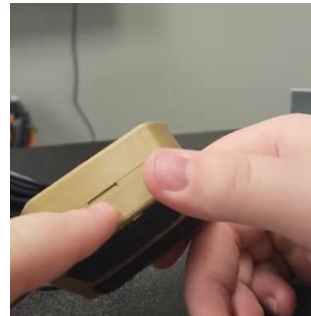
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1. We will replace the standard black small dome topper with a 3D printed topper.



2. First, locate the small slits on the side of the Spruce Joystick. This is where the top and bottom pieces of the Spruce Joystick join.



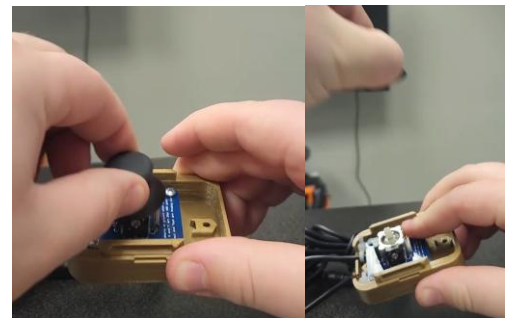
3. Hold the bottom half of the Spruce Joystick and press/pull the top half off.

Note: there are no screws holding the top and bottom half of the enclosure together, they just snap together.



4. You now have the two halves of the joystick enclosure separated. You can remove the standard topper by pulling straight up.

Note: the topper may have come off with the top half in step 3.



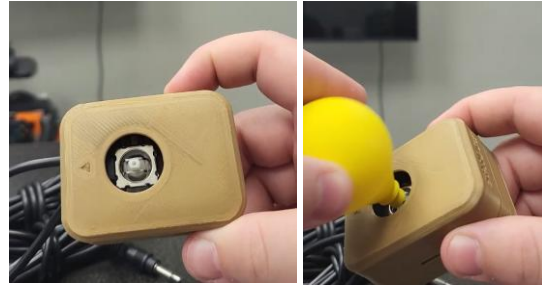
5. Line up the top half of the Spruce Joystick enclosure back onto the bottom half and press them together. Make sure the arrow on the top half is pointing towards the cables.



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6. Now, take your 3D printed topper and line up the rectangular slot on it with the white rectangular peg on the joystick and press them together.



7. You have now added a 3D printed topper. To swap 3D printed toppers, simply pull off the topper and press on a new one.

Note: You can put the standard topper back on by following these steps in reverse.



Specifications

Joystick Range of Motion	$\pm 30^\circ$
Joystick Range of Motion at Tip of Default Topper	± 15 mm
Joystick Activation Force at Tip of Default Topper	75 grams
Total Height with Default Topper	40 mm
Enclosure Height	31 mm
Enclosure Width	50 mm
Enclosure Length	65 mm
Weight (including cables)	108 grams

Cleaning

Wipe the outside of the device with a lukewarm, damp cloth. Do not use hot water on this device or the plastic housing may warp. Do not submerge the device.